

Project Report

TCP BIC vs TCP Reno

Siddharth Narayan Mishra
123CS0189

1. Introduction

TCP congestion control regulates data transmission to prevent network overload. This report compares two algorithms:

- **TCP Reno**: traditional AIMD-based approach
- **TCP BIC**: binary search-based high-speed congestion control

2. Objective

- Compare Reno and BIC behavior
- Analyze congestion window (cwnd) growth
- Demonstrate why BIC performs better

3. Methodology

A C++ simulation was used to model:

- Time steps representing RTTs
- Random packet loss
- Evolution of congestion window (cwnd)

TCP Reno:

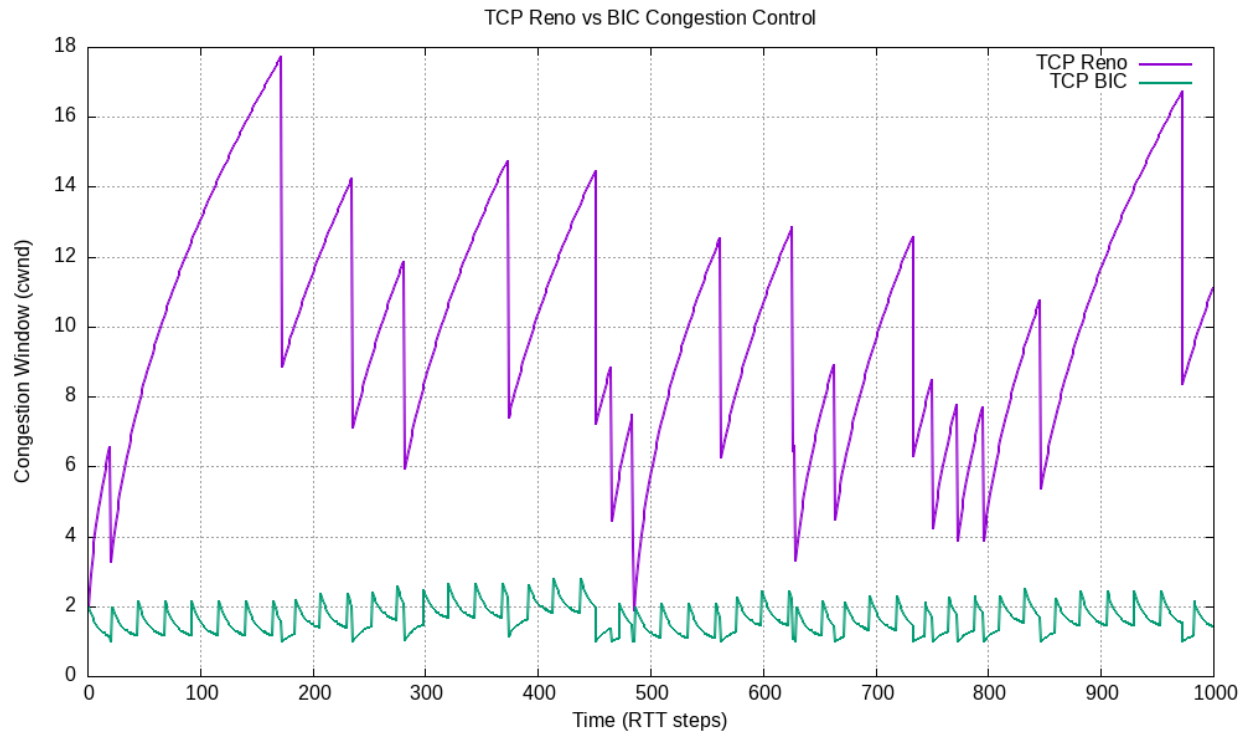
- Linear increase
- Halves cwnd on loss

TCP BIC:

- Uses binary search between min and max window
- Faster recovery and aggressive probing

4. Results

The graph below shows cwnd variation over time:



5. Analysis

- **Reno**
 - Slow linear growth
 - Poor recovery after loss
- **BIC**
 - Faster growth
 - Higher average cwnd
 - Better bandwidth utilization

6. Conclusion

TCP BIC outperforms TCP Reno in:

- Throughput
- Recovery speed
- Network utilization

Thus, BIC is more suitable for modern high-speed networks.